Chen Qian

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Applied Mathematics
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EDUCATION

- Ph.D. in Applied Mathematics — Sep. 2023 - June 2027 (expected)
  University of California, Davis — Davis, CA
  Research Advisor: Prof. Xiucai Ding, Department of Statistics, UC Davis
- M.S. in Statistics — Sep. 2021 - June 2023
  University of California, Davis — Davis, CA
- B.S. in Applied Statistics — Sep. 2016 - July 2020
  Complutense University of Madrid — Madrid, Spain

RESEARCH INTERESTS

- Statistical learning, inference, and reliability for dependent and sequential data.
- Representation learning and foundation models for dependent data.
- Biomedical time series applications: EEG, brain–computer interfaces (BCI), and neural speech decoding.

PUBLICATIONS

- **Qian, C.**, Ding, X., and Li, L. (2026). *Structural classification of locally stationary time series based on second-order characteristics.* **Journal of the Royal Statistical Society Series B: Statistical Methodology**, qkag083. DOI.
- Ding, X., **Qian, C.**, and Wang, J. (alphabetical order). *Structural Approximations for Multivariate Locally Stationary Time Series with Applications to Multichannel Signal Analysis.* **Submitted.**
- Ding, X., **Qian, C.**, and Zhou, Z. (alphabetical order). *On Estimation and Simultaneous Inferential Theory for the Autocorrelation Function of Locally Stationary Time Series.* **Work in progress.**

HONORS & AWARDS

- Conference Travel Award — 2026
  Conference on Statistical Foundations of AI, UC Santa Barbara (poster presentation forthcoming).
- L&S Graduate Summer Support Award, University of California, Davis — 2026
  Selected; declined due to COSMOS Teaching Fellow appointment.

SOFTWARE

[1] Ding, X., and **Qian, C.** (2023). **Sie2nts**: *Sieve Methods for Non-Stationary Time Series*. R package version 0.1.0. DOI; CRAN; User's guide.

[2] Ding, X., and **Qian, C.** (2023). **SIMle**: *Estimation and Inference for General Time Series Regression*. R package version 0.1.0. DOI; CRAN; User's guide.

RESEARCH EXPERIENCE

- **Structural Classification of Locally Stationary Time Series** Nov. 2024 - Apr. 2026
  Joint work with Xiucai Ding and Lexin Li; JRSS Series B, 2026
  - Developed a complete classification procedure for locally stationary time series, leveraging the autoregressive approximation framework for nonstationary time series to extract interpretable features of temporal dependence.
  - Designed and conducted simulation studies to evaluate parameter selection and robustness across different scenarios; performed real-data classification on the TUH Abnormal EEG Corpus, achieving 94.7% accuracy on test dataset using the Cz channel.
  - Collaborated with my advisor to establish feature consistency and asymptotic classification guarantees under class-separation conditions.
- **Structural Approximation of Multivariate Time Series and Multichannel EEG Classification** Jan. 2026 - Present
  Joint work with Xiucai Ding and Jeffrey Wang; Submitted
  - Implemented the classification procedure and developed strategies for dimension selection, norm selection, and hyperparameter tuning; conducted simulation studies across various dependence structures and class-balance settings.
  - Conducted comparisons with existing modern classification methods in both simulations and real-data applications.
  - Performed end-to-end multichannel EEG analysis on the Siena Scalp EEG Database, including preprocessing, parameter tuning, channel and norm selection, achieving 95.3% classification accuracy and an F1 score of 0.914.
- **Autocorrelation Estimation and Simultaneous Inference for Locally Stationary Time Series** June 2026 - Present
  Joint work with Xiucai Ding and Zhou Zhou; work in progress
  - Developed the constrained estimator for the time-varying autocorrelation function of locally stationary time series and constructed simultaneous confidence bands using the multiplier bootstrap procedure.
  - Designed and conducted simulation studies to evaluate estimation accuracy and confidence-band coverage.
  - Currently establishing uniform consistency of the estimator and asymptotic validity of the simultaneous confidence bands.

TEACHING EXPERIENCE

- Teaching Assistant 2023 - Present
  University of California, Davis Davis, CA
  Led discussion and office hours for courses in experiment design, data science, and calculus; prepared instructional materials and supported assessment.
- COSMOS summer program Teaching Fellow July 2026
  University of California, Davis Davis, CA
  Taught academically advanced high-school students core ideas in statistical modeling and exploratory data analysis. Mentored student teams in formulating research questions, selecting methods, interpreting results, and presenting technically sound conclusions.